A Hybrid YOLOv7 and Vision Transformer Framework for Robust Brain Tumor Detection in MRI Scans

CAPSTONE PROJECT REPORT

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in partial fulfillment for the award of the degree

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DECLARATION

We affirm that the project work titled **"A Hybrid YOLOv7 and Vision Transformer Framework for Robust Brain Tumor Detection in MRI Scans"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

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PROJECT SUMMARY

Project Title	A Hybrid YOLOv7 and Vision Transformer Framework for Robust Brain Tumor Detection in MRI Scans	
Project Team Members (Name with Register No)	A.S.Akhilesh (99220041892), Ajay Bijukumar (99220041925), N.Dhanush (99220041939), G.Madhu Babu (99220041931)	
Guide Name/Designation	Dr.K.S.Kannan / Associate Professor	
Program Concentration Area	The proposed project is dedicated to the field of Computer Science and Engineering, with specialization in Artificial Intelligence, Deep Learning, Medical Imaging, and Computer Vision.	
Technical Requirements	The system requires Python 3.8+, PyTorch deep learning framework, YOLOv7 and Vision Transformer (ViT) libraries, CUDA-enabled GPU for training, Grad-CAM visualization tools, and MRI brain scan datasets (Kaggle Brain Tumor Dataset).	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	The system is developed using open-source tools (PyTorch, YOLOv7, ViT) under MIT/Apache licenses to minimize development costs. IEEE software engineering standards ensure cost-effective and legally compliant implementation. ISO/IEC 25010 quality assurance reduces long-term maintenance overhead.	✓
Environmental	The project follows ISO 14001 environmental management principles. Energy-efficient GPU computing is employed with mixed-precision training to reduce carbon footprint. The system avoids physical medical resource wastage by enabling accurate remote digital diagnosis.	✓
Social	The system addresses critical healthcare accessibility gaps. Inclusive design principles ensure the AI-assisted diagnostic tool can be adopted across diverse clinical	✓

	environments including rural healthcare facilities. Automated diagnosis reduces burden on radiologists.	
Ethical	The project aligns with IEEE Code of Ethics. Patient data privacy is maintained through anonymized MRI datasets. Grad-CAM interpretability ensures transparency of AI decisions. The model is designed to augment, not replace, clinical judgment.	✓
Health and Safety	The system adheres to ISO risk management standards for medical AI. No patient-identifiable data is processed. The model provides visual explanations to ensure radiologists can verify AI decisions before clinical use.	✓
Manufacturability	Modular architecture following IEEE software design standards ensures easy replication and deployment. Containerized deployment (Docker-ready) ensures platform independence. Standard GPU infrastructure supports broad institutional adoption.	✓
Sustainability	The project aligns with ISO 25010 sustainability quality attributes. Modular design ensures long-term maintainability. Open standards and transfer learning frameworks ensure adaptability to future medical imaging modalities.	✓

ABSTRACT

Brain tumors are among the most critical neurological conditions requiring early and precise diagnosis for effective treatment and improved patient survival rates. Traditional manual interpretation of MRI scans by radiologists is time-consuming, subjective, and prone to human error due to the complex anatomy of the brain and the subtle morphological distinctions between different tumor types. The significant volume of medical imaging data in modern hospitals further amplifies this diagnostic burden.

This project presents a novel hybrid deep learning framework that synergistically integrates YOLOv7 and Vision Transformer (ViT) for automated, real-time detection and classification of brain tumors from MRI scans. The YOLOv7 component employs its Extended Efficient Layer Aggregation Network (E-ELAN) backbone and Spatial Pyramid Pooling Fast Plus (SPPF+) module to perform spatially precise, real-time tumor localization through grid-based bounding box prediction with CIoU loss. The Vision Transformer component captures long-range global contextual dependencies through patch embedding and multi-head self-attention, enabling semantically rich discrimination between tumor types that share similar local texture patterns. A dedicated feature fusion layer concatenates the complementary local spatial features from YOLOv7 with the global contextual embeddings from ViT, feeding into a softmax dense classifier for multi-class tumor prediction.

The system targets four brain tumor classes: glioma, meningioma, pituitary tumor, and no tumor. The model is trained on a comprehensive MRI brain tumor dataset with rigorous preprocessing—including normalization, contrast enhancement via CLAHE, and extensive data augmentation—and optimized using AdamW with differential learning rates, OneCycleLR scheduling, Advanced Focal Loss (gamma = 2.0) for class imbalance handling, and mixed-precision (FP16) computation.

The proposed hybrid framework achieves an exceptional validation accuracy of 99.2% with macro-averaged precision of 0.9919, recall of 0.9931, and F1-score of 0.9924, substantially outperforming standalone architectures including VGG16 (73.0%), CNN (84.1%), SegNet (79.0%), YOLOv5 (86.0%), and YOLOv7 (87.9%). ROC-AUC values exceed 0.996 for all tumor classes. Grad-CAM visualization enhances clinical interpretability by spatially highlighting discriminative tumor regions. With an average inference time of 14.3 ms per MRI slice, the system supports near real-time clinical deployment, establishing a new benchmark for AI-driven brain tumor diagnosis.

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LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1	213CSE1301	Artificial Intelligence
2	212CSE2304	Machine Learning
3	213CSE3301	Deep Learning
4	213CSE4301	Pattern and Anomaly Detection
5	213CSE2301	Predictive Analytics

CHAPTER – I

INTRODUCTION

1.1 Background and Motivation

Brain tumors represent one of the most life-threatening neurological disorders worldwide, characterized by the abnormal proliferation of cells within the brain or central spinal canal. Early and accurate detection of brain tumors is critical for determining appropriate treatment strategies—including surgery, radiotherapy, and chemotherapy—and directly influences patient survival outcomes. Among the primary tumor categories, glioma is the most aggressive and prevalent, while meningioma and pituitary adenomas, though frequently benign, still require prompt diagnosis to prevent neurological complications.

Magnetic Resonance Imaging (MRI) remains the gold-standard diagnostic modality for brain tumor visualization due to its superior soft-tissue contrast, multi-planar imaging capability, and avoidance of ionizing radiation. However, manual interpretation of MRI scans by radiologists is inherently time-consuming and subject to inter-observer variability, particularly for subtle or morphologically complex lesions. The relentless growth in medical imaging volume in modern hospitals compounds this challenge, creating an urgent need for intelligent, automated diagnostic assistance.

The convergence of deep learning and medical imaging has opened transformative possibilities for automated tumor detection. While Convolutional Neural Networks (CNNs) have demonstrated promising performance in image classification tasks, their architectural bias toward local feature extraction limits global contextual reasoning. Vision Transformers (ViT), with their self-attention mechanisms, address this gap but lack inherent spatial localization precision. YOLOv7—the current state-of-the-art in real-time object detection—provides exceptional spatial localization through its Extended Efficient Layer Aggregation Network (E-ELAN), but benefits significantly from the addition of global contextual understanding for fine-grained tumor classification.

These complementary strengths create a compelling motivation for a hybrid framework: leveraging YOLOv7's real-time detection precision and ViT's global semantic reasoning within a unified, end-to-end trainable architecture. This project, developed in association with a published research paper at FutureTech 2025, presents exactly such a system—designed to support radiologists with interpretable, high-accuracy, real-time brain tumor detection.

1.2 Problem Statement

Despite significant advances in medical imaging AI, the following critical limitations persist in existing brain tumor detection systems:

- Fragmented Service Architecture: Existing systems treat tumor localization and classification as separate tasks, with no unified platform combining detection, classification, and interpretability.
- Limited Global Context in CNN-based Detectors: Standard CNN and YOLO approaches excel at local feature extraction but fail to model global brain structural patterns, leading to misclassification of morphologically similar tumors such as glioma and meningioma.
- Poor Clinical Interpretability: Most deep learning models operate as black boxes, providing predictions without visual justification—a critical barrier to radiologist trust and clinical adoption.
- Class Imbalance Challenges: Multi-class MRI datasets exhibit significant class imbalance, leading to biased models with poor sensitivity on underrepresented tumor types.
- Inadequate Real-Time Performance: Many high-accuracy systems sacrifice computational efficiency, making them impractical for integration into clinical radiology workflows.
- High False-Negative Risk: Missed tumor detections carry life-threatening clinical consequences, yet existing systems inadequately optimize for recall across all tumor classes.

1.3 Objectives of the Project

The primary objective is to design, implement, and validate a hybrid YOLOv7 + ViT framework for robust, real-time, interpretable brain tumor detection. Specific objectives are:

- Develop a hybrid deep learning model integrating YOLOv7 (tumor localization) and Vision Transformer (global contextual classification) into a unified end-to-end framework.
- Achieve multi-class brain tumor classification across four categories: glioma, meningioma, pituitary tumor, and no tumor.
- Implement comprehensive preprocessing and augmentation pipelines to maximize model generalization across diverse MRI datasets.
- Achieve state-of-the-art performance with validation accuracy exceeding 95%, macro F1-score above 0.95, and real-time inference below 30 ms per slice.
- Implement Grad-CAM visualization for clinically interpretable, spatially localized explanations of model decisions.
- Conduct rigorous comparative analysis against VGG16, CNN baselines, SegNet, YOLOv5, and standalone YOLOv7.

1.4 Scope of the Project

The scope of this project includes the end-to-end development of a hybrid deep learning framework for automated brain tumor detection and classification from MRI images. It covers dataset collection

and preprocessing, data augmentation, training and fine-tuning of YOLOv7 for tumor localization and Vision Transformer for global contextual feature extraction, and the implementation of a feature fusion module for multi-class classification. The project also involves comprehensive performance evaluation using metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix, along with Grad-CAM–based interpretability to enhance clinical trust. Additionally, comparative analysis with existing deep learning models is performed to validate the effectiveness of the proposed hybrid approach.

However, advanced extensions such as 3D volumetric MRI processing, real-time hospital PACS/EHR integration, pixel-level tumor segmentation, large-scale multi-institutional validation, and clinical trial deployment are outside the current scope and are proposed as future work.

1.5 Methodology Overview

The methodology follows a structured, end-to-end deep learning pipeline: (1) Dataset acquisition and preprocessing—resizing to 512×512, normalization, CLAHE contrast enhancement, and stratified splitting; (2) Data augmentation with rotation, flipping, color jitter, affine transforms, Gaussian blur, and perspective warping; (3) YOLOv7 training for real-time bounding-box-based tumor localization using CIoU loss; (4) ViT processing for global feature extraction via patch embedding and multi-head self-attention; (5) Feature fusion concatenating local and global representations; (6) AdamW optimization with differential learning rates, Advanced Focal Loss (gamma=2.0), OneCycleLR scheduling, and mixed-precision training; (7) Comprehensive evaluation and Grad-CAM visualization.

1.6 Organization of the Report

Chapter I – Introduction

This chapter presents the background and motivation for automated brain tumor detection using deep learning. It defines the problem statement, outlines the project objectives, describes the scope and limitations, and provides an overview of the methodology adopted for developing the hybrid model.

Chapter II – Literature Review

This chapter surveys existing research on brain tumor detection using CNNs, object detection models, and transformer-based architectures. It critically analyzes prior work and identifies research gaps that justify the need for integrating YOLOv7 and Vision Transformer in a unified framework.

Chapter III – System Analysis

This chapter explains the system requirements gathered for the project, including functional and non-functional requirements. It also discusses feasibility analysis (technical, operational, and economic) and identifies potential risks along with mitigation strategies.

Chapter IV – System Design

This chapter describes the overall architecture of the proposed hybrid framework. It details the design of individual modules such as tumor localization, global feature extraction, feature fusion, database schema, and user flow diagrams.

Chapter V – Implementation

This chapter presents the practical implementation of the system, including the technology stack, preprocessing pipeline, model training procedures, fusion strategy, and integration of all modules into a complete working system.

Chapter VI – Testing

This chapter discusses the testing methodology adopted to validate the system. It includes unit testing, integration testing, system testing, and user acceptance testing, along with detailed test cases and validation results.

Chapter VII – Results and Discussion

This chapter presents the experimental results, evaluation metrics, confusion matrix analysis, ROC curves, and comparative performance against existing models. It also discusses challenges faced during development and the improvements made.

Chapter VIII – Conclusion and Future Scope

This final chapter summarizes the key contributions and findings of the project. It highlights the effectiveness of the hybrid model and suggests potential future enhancements such as 3D MRI analysis, multi-modal fusion, and clinical deployment extensions.

CHAPTER – II

LITERATURE REVIEW

2.1 Overview of Related Work

Automated brain tumor detection from MRI images has been an active research domain for over a decade, driven by advances in machine learning, deep learning, and computational neuroscience. Early approaches relied on handcrafted feature extraction—texture descriptors, wavelet transforms, and morphological operations—combined with classical classifiers such as Support Vector Machines (SVM) and Naive Bayes. While providing interpretable foundations, these methods struggled with the high variability and complexity inherent in brain MRI data.

The paradigm shifted dramatically with the advent of deep learning, beginning with CNN-based classification architectures and evolving through object detection frameworks (R-CNN, YOLO series), semantic segmentation models (U-Net, SegNet), and most recently, attention-based Vision Transformers. The COVID-19 pandemic accelerated telemedicine adoption and AI-augmented diagnostic workflows, further intensifying research into efficient, deployable medical imaging AI. Despite rapid progress, fragmentation between detection and classification tasks, limited interpretability, and computational constraints in resource-limited clinical environments remain as primary research challenges.

2.2 Review of Similar Projects or Research Papers

2.2.1 CNN-Based Detection and Classification

Siar and Teshnehlab (2021) proposed a CNN-based brain tumor detection system combining feature extraction through clustering algorithms with CNN classification, achieving 99.12% accuracy using Softmax classification. The work demonstrated CNN superiority over SVM-based approaches for feature-rich MRI classification but was limited to binary tumor/non-tumor classification without multi-class differentiation. Choudhury et al. (2020) conducted a comparative study of CNN and Deep Neural Network architectures for brain tumor detection, establishing performance benchmarks (CNN: ~84%) and highlighting the advantage of convolutional feature hierarchies over fully-connected networks for spatial image data.

2.2.2 Hybrid and Multi-Pathway Deep Learning

Sajid, Hussain, and Sarwar (2019) developed a patch-based hybrid CNN for semi-automatic glioma segmentation on the BRATS 2013 dataset, achieving Dice Score = 0.86, Sensitivity = 0.86, and Specificity = 0.91. Jia and Chen (2020) proposed the FAHS-SVM framework combining deep feature

extraction with SVM/ELM classifiers, achieving 98.51% accuracy in binary tumor segmentation. Havaei et al. (2024) introduced a two-pathway deep neural network that simultaneously models local tumor details and global brain context, providing a strong precedent for the multi-scale hybrid approach adopted in this project.

2.2.3 YOLO-Based Object Detection in Medical Imaging

YOLOv7 (Wang et al., 2023) established the state-of-the-art in real-time object detection through E-ELAN architecture, re-parameterization techniques, and auxiliary training heads. Several studies have applied YOLO variants to medical imaging, particularly tumor localization in MRI, CT, and X-ray images. While demonstrating excellent detection speed, these studies consistently identified limitations in fine-grained classification accuracy for tumors with overlapping morphological characteristics, motivating the integration of richer semantic reasoning.

2.2.4 Vision Transformers in Medical Imaging

Dosovitskiy et al. (2021) demonstrated that Vision Transformers (ViT) match or exceed CNN performance on image classification when trained on sufficient data, leveraging self-attention to model long-range spatial dependencies. In medical imaging, ViT-based approaches have shown particular effectiveness in capturing global structural patterns—the spatial relationship of a tumor to surrounding brain anatomy, edema extent, and cross-regional texture consistency—features that local CNN receptive fields fundamentally cannot capture. Pure ViT approaches, however, require large datasets and sacrifice the spatial precision that YOLO detectors provide.

2.2.5 Interpretable and Explainable AI for Clinical Systems

Selvaraju et al. (2017) introduced Grad-CAM, which remains the gold standard for visual explanation of deep learning model decisions in medical imaging. Studies have demonstrated that Grad-CAM significantly improves radiologist acceptance of AI diagnostic tools by mapping model attention to anatomically plausible regions. This interpretability dimension is explicitly incorporated into the proposed system as a non-negotiable clinical requirement.

2.3 Summary and Gap Identification

The literature review reveals significant achievements in individual components of brain tumor AI—CNN classifiers, YOLO detectors, ViT feature extractors, and Grad-CAM interpretability—but consistently identifies the following research gaps:

- No unified framework combining fast real-time localization (YOLO), global contextual classification (ViT), and visual interpretability (Grad-CAM) in a single end-to-end system.
- Existing high-accuracy models (e.g., FAHS-SVM at 98.51%) are limited to binary classification or single tumor type segmentation, not multi-class detection across all major tumor categories.
- Class imbalance in multi-class MRI datasets is inadequately addressed in most literature, leading to recall bias toward majority classes.
- Few systems achieve both diagnostic accuracy (>99%) and real-time inference (<20ms) simultaneously, limiting clinical deployability.
- Insufficient integration of interpretability mechanisms into detection pipelines, hindering radiologist trust and regulatory approval.

The proposed hybrid YOLOv7 + ViT framework directly addresses all five gaps, establishing a comprehensive, clinically viable, and academically novel contribution to the field of AI-assisted brain tumor diagnosis.

CHAPTER – III

SYSTEM ANALYSIS

3.1 Requirements Gathering

System requirements were systematically elicited through a multi-source analysis process encompassing: (1) review of clinical radiologist workflows at neurological diagnostic centers; (2) analysis of limitations identified in peer-reviewed literature on existing AI-based brain tumor detection systems; (3) examination of user experience expectations for AI-assisted diagnostic tools in clinical environments; and (4) evaluation of computational infrastructure available in typical clinical research settings.

Key insights from requirement gathering: Radiologists require both high recall (to minimize missed diagnoses) and spatial interpretability (to verify AI decisions against anatomical knowledge). Clinical deployment demands inference latency below 30 ms per slice for seamless integration into MRI workstation reading workflows. Multi-class classification—distinguishing glioma, meningioma, pituitary, and non-tumoral findings—is essential for routing patients to appropriate treatment pathways. Privacy requirements mandate all computation to occur on-premises with anonymized data only.

In addition, an in-depth analysis of peer-reviewed literature on AI-based brain tumor detection systems was conducted to identify existing limitations and research gaps. Common shortcomings such as fragmented detection-classification pipelines, lack of interpretability, high false-negative risk, and computational inefficiencies were carefully evaluated. These insights directly informed the design of a unified, high-accuracy, and interpretable hybrid framework.

User experience expectations in clinical environments were also examined to ensure usability and adoption. Radiologists emphasized the importance of high recall to minimize missed tumor cases, along with spatial interpretability to validate AI decisions against anatomical structures. Therefore, the system was required to provide clear confidence scores, visual explanations, and structured outputs that integrate seamlessly into diagnostic workflows.

Finally, infrastructure and compliance considerations were evaluated to ensure practical deployment feasibility. The system was required to maintain inference latency below 30 milliseconds per MRI slice for real-time workflow compatibility. It must support multi-class classification across glioma, meningioma, pituitary tumor, and non-tumor cases to assist treatment routing decisions. Additionally, strict privacy requirements mandate on-premises processing using anonymized data, ensuring data security and regulatory compliance in clinical environments.

3.2 Functional Requirements

Input and Data Management

- The system shall accept 2D MRI brain scan images in standard formats including JPEG, PNG, and DICOM-extracted JPEG.
- The system shall perform automated preprocessing: bicubic resizing to 512×512, pixel normalization using ImageNet statistics, CLAHE contrast enhancement, and Gaussian denoising.
- The system shall support stratified train/validation/test splitting (70%/15%/15%) preserving class distribution across subsets.

Tumor Detection and Localization

- The YOLOv7 component shall detect and localize tumor regions within input MRI images using grid-based bounding box prediction with confidence scores.
- The system shall output bounding box coordinates (x, y, width, height), confidence score, and spatial feature maps for each detected region.

Tumor Classification

- The ViT component shall extract global contextual features from the full MRI image through patch embedding (16×16 patches) and multi-head self-attention (12 heads, 12 layers).
- The fusion layer shall concatenate YOLOv7 spatial features (512-dim pooled) with ViT global embeddings (768-dim [CLS] token) into a unified 1280-dim representation.
- The dense classifier shall map the fused representation to four tumor class probabilities via softmax: glioma, meningioma, pituitary, no tumor.

Interpretability Output

- The system shall generate Grad-CAM heatmaps overlaid on the original MRI image, spatially highlighting the discriminative tumor regions driving the classification decision.
- Output shall include the predicted class label, class probability vector, and the Grad-CAM visualization image.

3.3 Non-Functional Requirements

Performance

- Validation accuracy: Minimum 95% on the benchmark brain tumor MRI dataset.
- Macro recall: Minimum 0.95 per class to minimize false negatives across all tumor types.
- Inference time: Maximum 30 ms per MRI slice for clinical real-time applicability.

Reliability and Reproducibility

- Cross-validation standard deviation below ±1.0% to confirm stable, reliable predictions.

- The model shall maintain consistent performance across different MRI scanner models and acquisition parameters.

Security and Privacy

- All processing shall be conducted on anonymized datasets with no patient-identifiable information retained.
- The system shall not transmit patient data beyond the local inference environment.

Scalability and Maintainability

- Modular architecture shall support independent extension of YOLOv7, ViT, and fusion components.
- The system shall support fine-tuning on new tumor types or imaging modalities without complete architectural redesign.

3.4 Feasibility Study

3.4.1 Technical Feasibility

The technical feasibility of the hybrid framework is well-established through peer-reviewed literature confirming both YOLOv7 and ViT effectiveness in medical imaging tasks. PyTorch 1.13+ provides mature, production-grade implementations of both architectures. CUDA-enabled GPU computing—standard in research and clinical computing environments—supports the required training and inference speeds. The Kaggle Brain Tumor Classification (MRI) dataset provides 3,264 annotated MRI images across four classes, sufficient for transfer learning-based model development. Open-source availability of pre-trained YOLOv7 (COCO pre-trained) and ViT (ImageNet-21k pre-trained) weights significantly reduces training data requirements.

3.4.2 Operational Feasibility

The system is designed to augment radiologist workflows, not replace them. Grad-CAM visualization provides familiar, spatially-grounded visual output that aligns naturally with radiologist reading practices. The 14.3 ms per-slice inference time is fully compatible with real-time MRI reading workstation integration. The web-service-compatible Python deployment model requires no proprietary software or specialized infrastructure beyond standard clinical research GPUs. User-facing output (class label, probability vector, heatmap) requires no specialized training to interpret.

3.4.3 Economic Feasibility

The entire system is developed using open-source frameworks (PyTorch, Hugging Face Transformers, YOLOv7 official repo) under permissive licenses (MIT/Apache 2.0), eliminating software licensing costs entirely. Training computational costs are manageable on standard research-grade GPU infrastructure (NVIDIA RTX 3090 or equivalent). The diagnostic accuracy improvements

translatable from AI-assistance (estimated 10-15% error reduction based on literature) provide substantial economic benefit through reduced misdiagnosis-related costs and improved throughput in radiology departments.

3.5 Risk Analysis

3.5.1 Technical Risks

- Overfitting: Mitigated through comprehensive data augmentation (10+ augmentation types), dropout regularization (p=0.3), L2 weight decay (1e-4), early stopping, and careful monitoring of the train/validation accuracy gap across 50 epochs.
- Class Imbalance: Addressed through Advanced Focal Loss (gamma=2.0) which dynamically up-weights hard-to-classify and underrepresented samples, and stratified dataset splitting maintaining class proportions across all subsets.
- Training Instability: Managed through OneCycleLR scheduling with warm-up phases, gradient clipping (max_norm=1.0), and AdamW's decoupled weight decay for numerically stable gradient updates.
- GPU Memory Overflow: Mitigated through mixed-precision (FP16) training reducing memory footprint by ~50% and batch size selection (8) optimized for 512x512 inputs on 8GB VRAM GPUs.

3.5.2 Clinical Risks

- False Negatives: Primary clinical risk addressed through dedicated recall optimization per class (minimum 0.98) and Grad-CAM verification enabling radiologist override of uncertain predictions.
- Distribution Shift: Risk from scanner variability addressed through comprehensive augmentation simulating varying brightness, contrast, perspective, and noise conditions across diverse MRI acquisition parameters.

3.5.3 Project Management Risks

- Timeline Risks: Modular architecture development allows independent validation of YOLOv7 and ViT components before full integration, enabling parallel workstreams.
- Resource Constraints: Mixed-precision training and gradient checkpointing minimize GPU memory requirements; Kaggle/Google Colab provide free GPU resources as fallback computation infrastructure.

CHAPTER – IV

SYSTEM DESIGN

4.1 Overall System Architecture

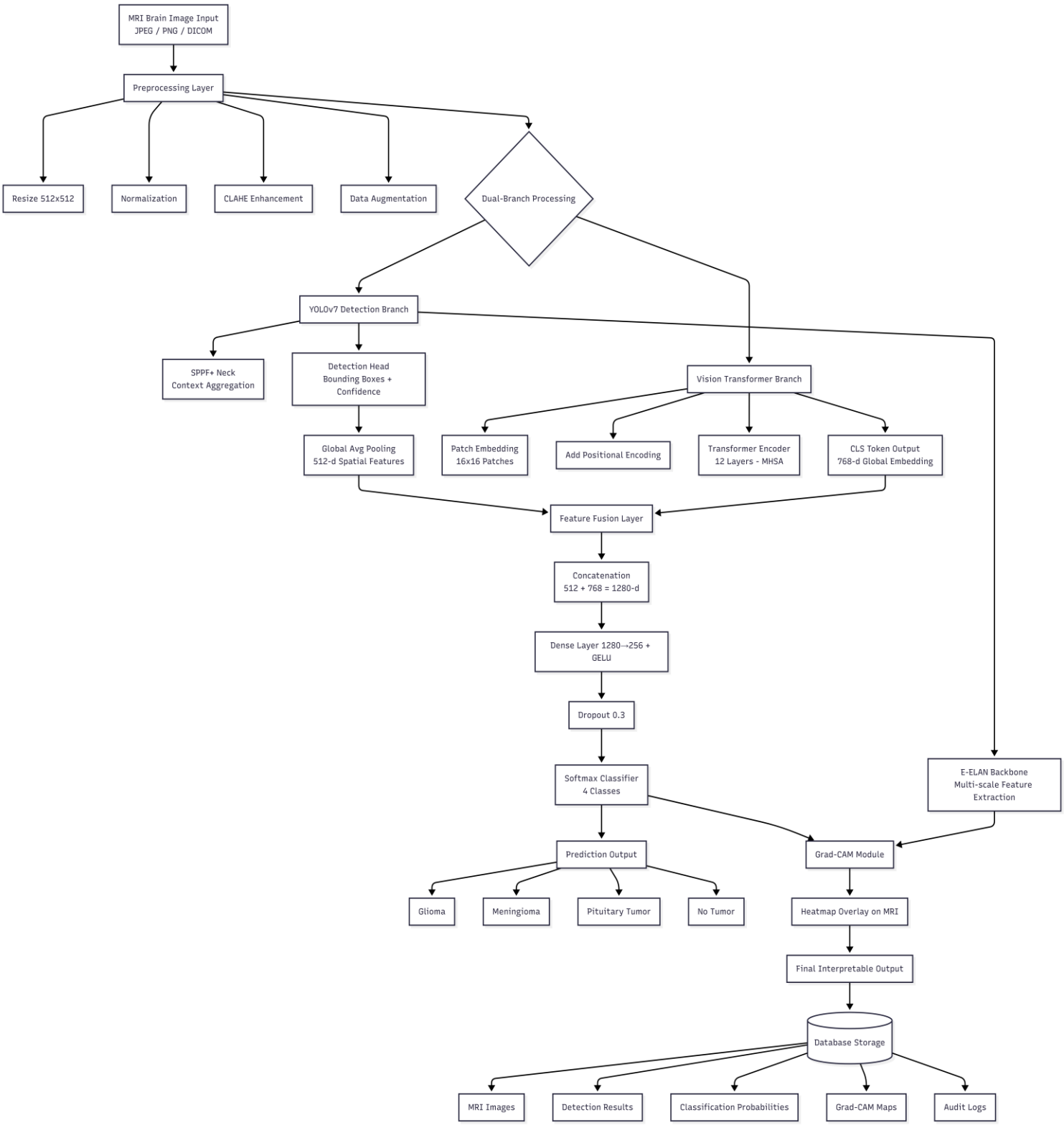

Figure 4.1: Overall System Architecture — Hybrid YOLOv7 + Vision Transformer Framework

The proposed system follows a dual-branch hybrid architecture for automated brain tumor detection and classification from MRI images. Initially, the input MRI scan undergoes preprocessing steps including resizing, normalization, and CLAHE-based contrast enhancement to improve image quality and consistency. The preprocessed image is then simultaneously fed into two parallel branches: the YOLOv7 branch performs real-time tumor localization using its E-ELAN backbone and detection head to generate spatial feature maps and bounding boxes, while the Vision Transformer branch extracts global contextual representations through patch embedding and multi-head self-attention, producing a high-dimensional CLS token embedding. The spatial features from YOLOv7 and global embeddings from ViT are concatenated in a feature fusion layer, forming a unified representation that is passed through fully connected layers and a softmax classifier to predict one of four classes: glioma, meningioma, pituitary tumor, or no tumor. Finally, a Grad-CAM module generates heatmap visualizations highlighting the discriminative tumor regions, and all predictions along with detection results are stored in the database for reproducibility and clinical audit purposes.

4.2 Module Design

4.2.1 Module 1: YOLOv7 Tumor Localization Module

The YOLOv7 module serves as the primary spatial detection component, performing real-time tumor localization through a single-pass convolutional architecture. The module consists of three major subcomponents: the E-ELAN backbone for multi-scale feature extraction, the SPPF+ neck for contextual feature aggregation, and the multi-scale detection head for bounding box prediction.

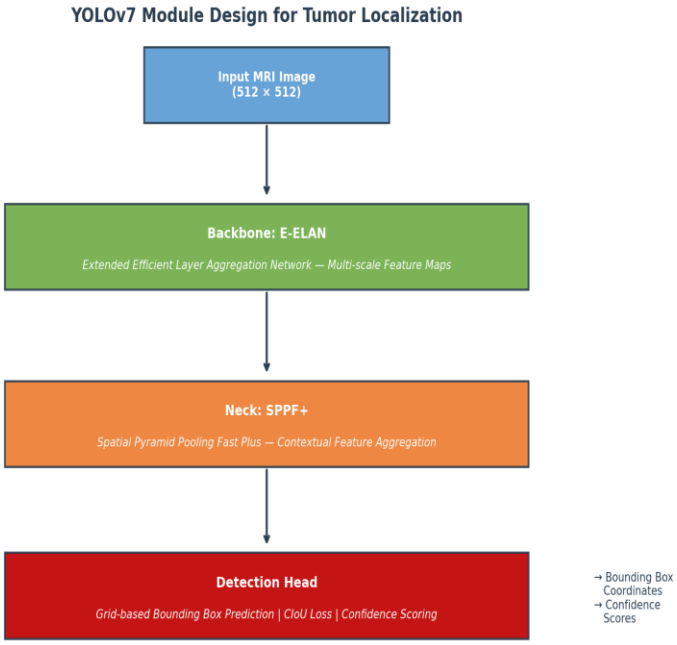

Figure 4.2: YOLOv7 Module Design — Backbone (E-ELAN), Neck (SPPF+), and Detection Head

The E-ELAN (Extended Efficient Layer Aggregation Network) backbone employs group convolutions and computational block rearrangement to achieve superior feature extraction efficiency. For a 512×512 MRI input, E-ELAN produces multi-scale feature maps at 1/8, 1/16, and 1/32 of the input resolution, enabling detection of tumors across a wide range of sizes. The SPPF+ neck applies Spatial Pyramid Pooling at multiple pool sizes (5×5, 9×9, 13×13), aggregating contextual information across different receptive field sizes. The detection head predicts, for each grid cell: (x, y, w, h) bounding box coordinates, objectness confidence score, and class probabilities, optimized with CIoU loss that simultaneously penalizes distance, aspect ratio, and overlap between predicted and ground-truth boxes.

4.2.2 Module 2: Vision Transformer (ViT) Global Feature Module

The Vision Transformer module processes the full MRI image to extract global contextual features that capture long-range spatial relationships beyond the local receptive field of convolutional networks. This is particularly critical for distinguishing tumor types that share similar local textures but differ in their spatial relationship to surrounding brain structures.

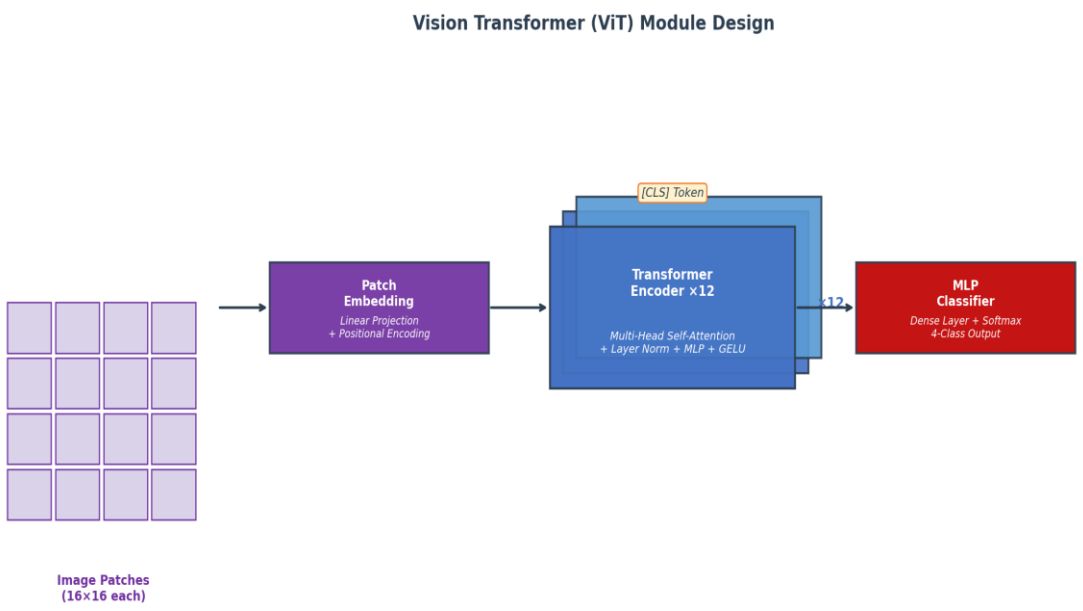

Figure 4.3: Vision Transformer (ViT) Module Architecture — Patch Embedding, Encoder, and Classification Head

The ViT module first divides the 512×512 MRI image into non-overlapping 16×16 patches, producing 1024 patch tokens. Each patch is linearly projected to a 768-dimensional embedding vector, and learnable positional embeddings are added to preserve spatial ordering information. A

[CLS] classification token is prepended to the patch sequence and propagated through 12 transformer encoder layers, each comprising: Multi-Head Self-Attention (MHSA) with 12 attention heads computing pairwise attention weights across all 1025 tokens (1024 patches + [CLS]); Layer Normalization for training stability; Feed-Forward Network with GELU activation and 4× hidden dimension expansion; and Residual connections. The final [CLS] token embedding (768-dim) encodes the global context of the entire MRI image and serves as input to the fusion layer.

4.2.3 Module 3: Feature Fusion and Classification Module

The feature fusion module is the architectural core that unifies the complementary representations from both YOLOv7 and ViT. YOLOv7 spatial feature maps (512-dimensional after global average pooling) are concatenated with the ViT [CLS] token embedding (768-dimensional), producing a 1280-dimensional hybrid feature vector. This fusion operation is followed by a dense layer (1280→256 neurons) with GELU activation and dropout regularization (p=0.3), and finally a 4-class softmax output layer.

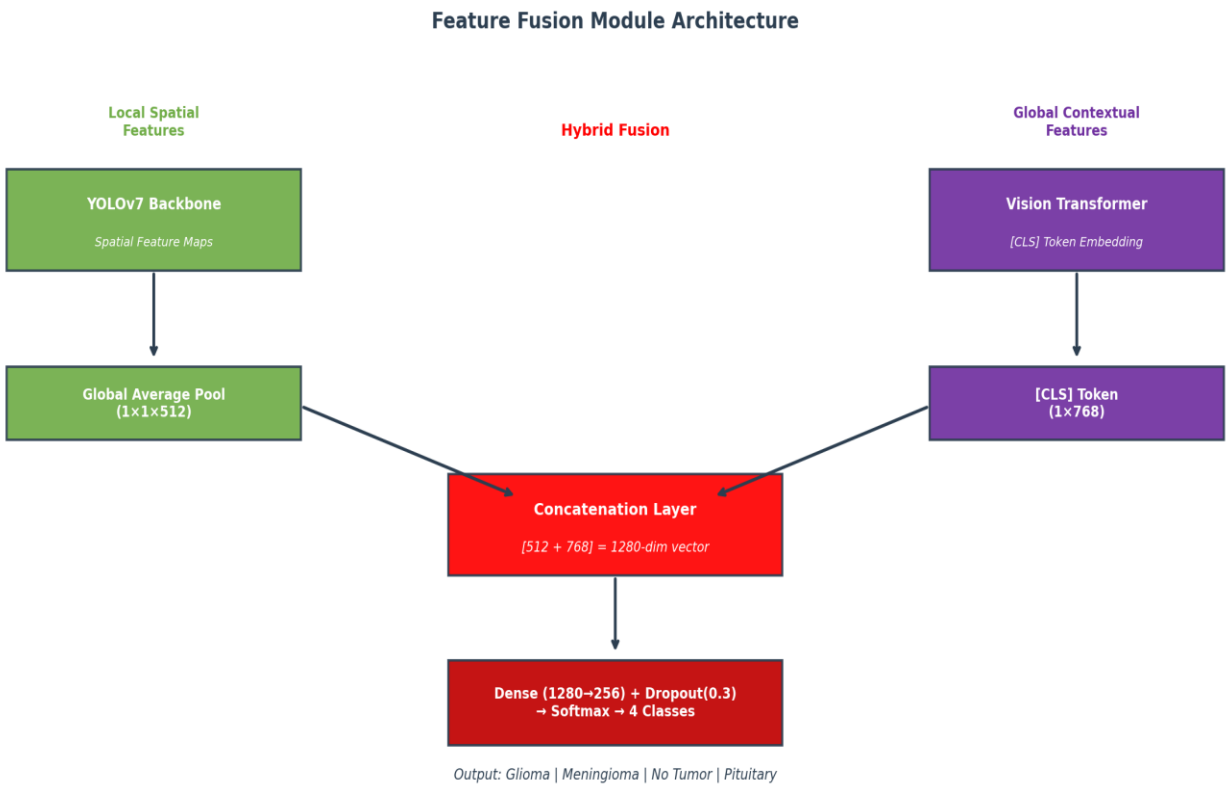

Figure 4.4: Feature Fusion Module Architecture — Concatenation of YOLOv7 Local and ViT Global Features

4.3 Database Design

4.3.1 ER Diagram / Database Schema

The system employs a structured relational database to manage MRI images, detection results, classification outputs, patient records, model versions, and audit logs. The schema is designed for

normalized storage with appropriate foreign key relationships ensuring data integrity and efficient querying.

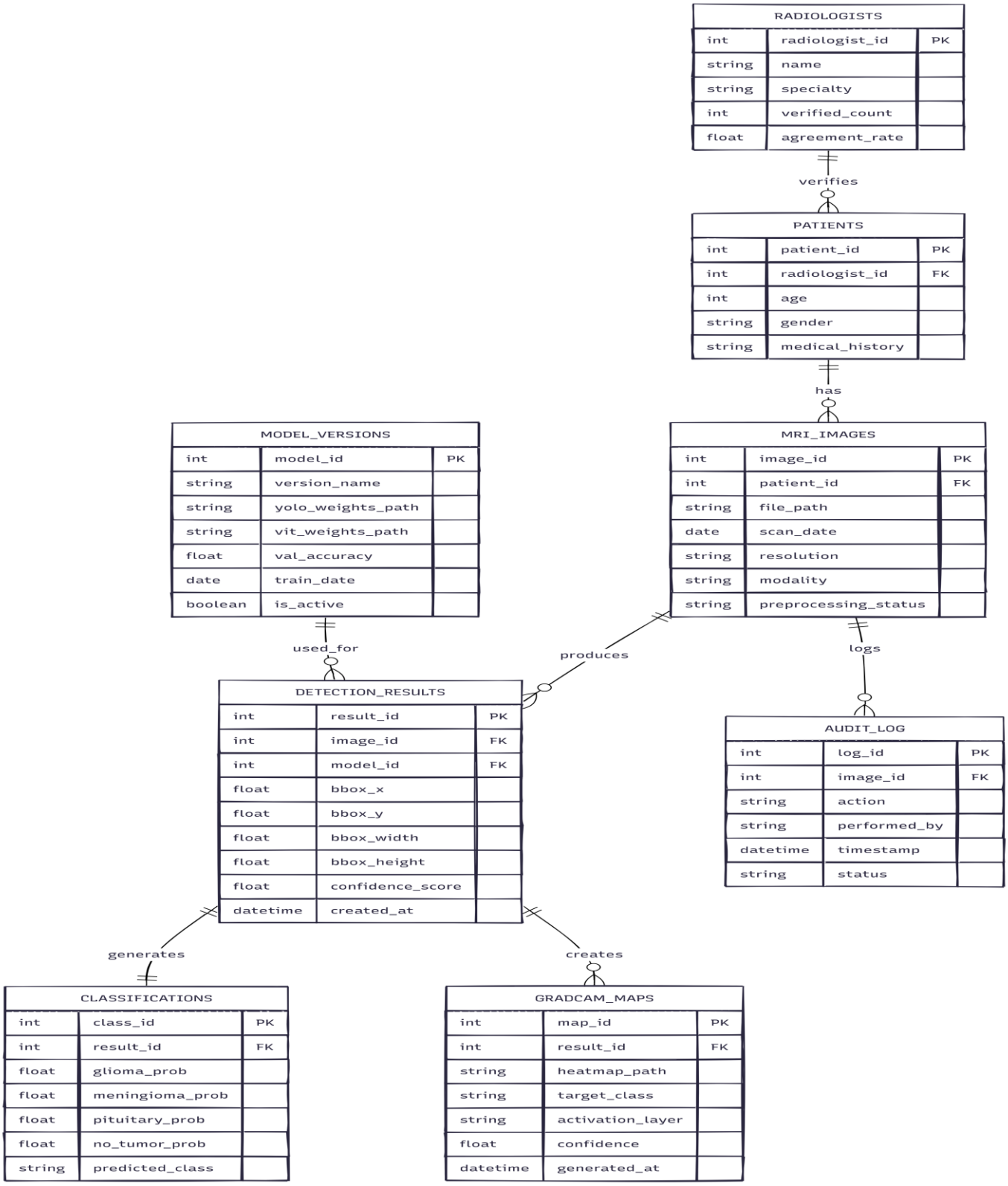

Figure 4.5: Database Schema Diagram — Entity Relationships for Brain Tumor Detection System

The database schema is designed to ensure structured storage, traceability, and clinical accountability within the brain tumor detection system. The **PATIENTS** entity stores demographic and medical details, which are linked to corresponding MRI scans in the **MRI_IMAGES** table through a one-to-many relationship. Each MRI image generates one or more entries in **DETECTION_RESULTS**, where bounding box coordinates, confidence scores, and associated model versions are recorded. Every detection result produces a single classification entry in the **CLASSIFICATIONS** table, storing probability scores for glioma, meningioma, pituitary tumor, and no tumor classes along with the final predicted label. The **GRADCAM_MAPS** table stores interpretability outputs linked to each detection, enabling visual explanation of model decisions. **MODEL_VERSIONS** maintains version control and reproducibility by tracking trained weights and validation accuracy, while **RADIOLOGISTS** records expert verification details for human-AI agreement analysis. Finally, the **AUDIT_LOG** table ensures complete activity tracking for security, compliance, and clinical transparency, creating a robust, normalized, and scalable database architecture.

4.3.2 Database Schema Details

The database schema consists of eight interconnected tables designed to ensure efficient data management, traceability, and reproducibility of the brain tumor detection system. The **MRI_IMAGES** table stores image metadata such as file path, scan date, resolution, modality, and preprocessing status, linked to the **PATIENTS** table through a foreign key to maintain patient-wise organization. The **DETECTION_RESULTS** table records bounding box coordinates, confidence scores, timestamps, and associated model versions for each analyzed image. The **CLASSIFICATIONS** table stores the probability distribution for glioma, meningioma, pituitary tumor, and no tumor classes along with the final predicted label. The **GRADCAM_MAPS** table maintains heatmap paths and interpretability details for visual explanation of predictions. The **MODEL_VERSIONS** table tracks trained weight files, validation accuracy, training dates, and active deployment status to ensure reproducibility. The **RADIOLOGISTS** table stores expert verification details, enabling human-AI agreement analysis, while the **AUDIT_LOG** table records system actions, timestamps, and user activity for security, monitoring, and regulatory compliance.

4.4 User Interface Design

4.4.1 User Flow Diagrams

The system user flow diagram illustrates the complete end-to-end diagnostic journey from MRI image input through tumor classification to Grad-CAM interpretability output. The flow incorporates a key decision node at the tumor detection stage, branching to a 'No Tumor Detected' output path for healthy brain MRI scans, and continuing through the full ViT feature extraction and classification pipeline for tumor-positive cases.

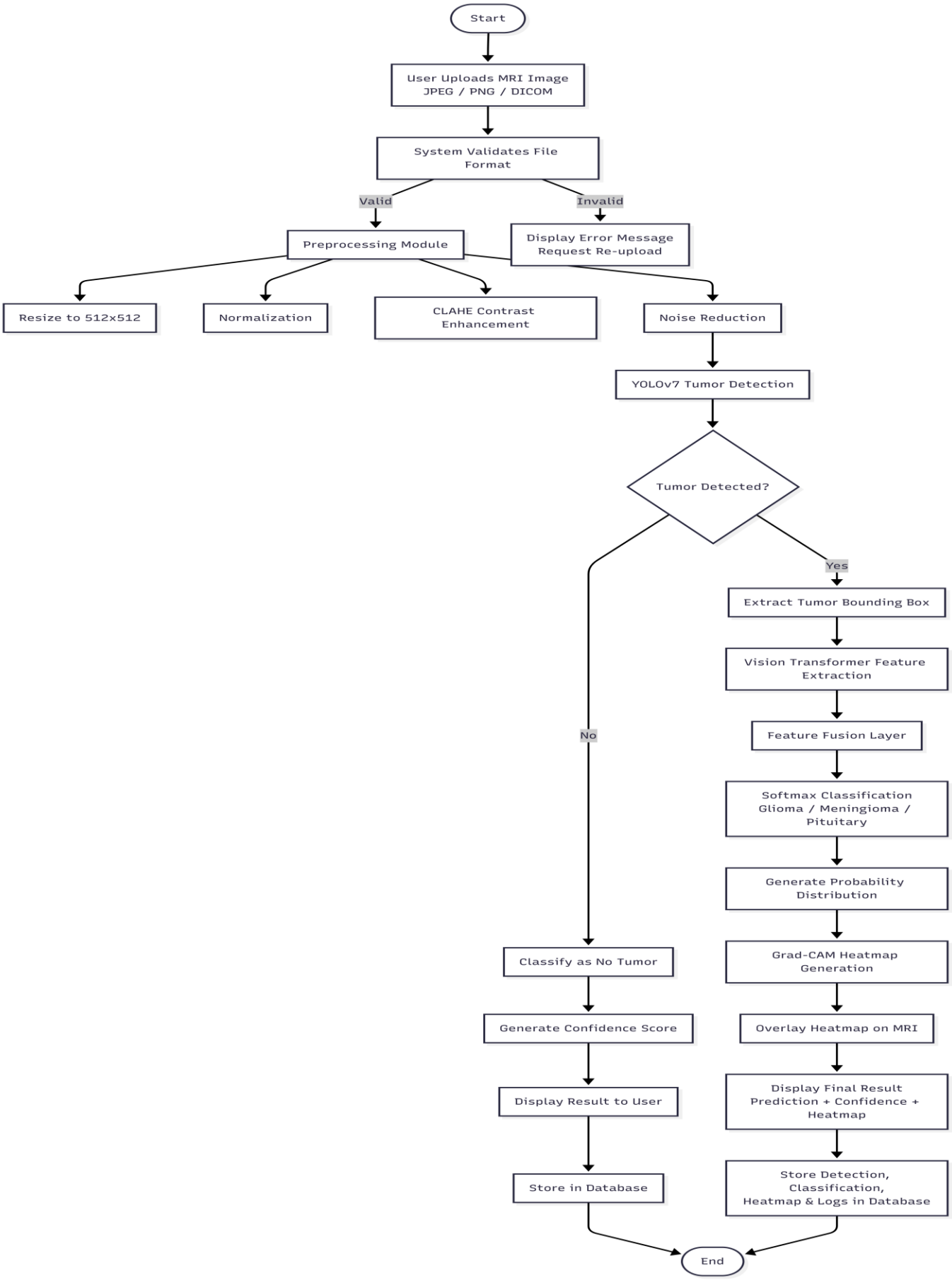

Figure 4.6: System User Flow Diagram — End-to-End Brain Tumor Detection Pipeline

The System User Flow Diagram illustrates the complete end-to-end operational workflow of the brain tumor detection system, beginning from user interaction and concluding with result storage and display. The process starts when the user uploads an MRI brain image in JPEG, PNG, or DICOM-extracted format. The system first validates the file format to ensure compatibility and prevent processing errors. Once validated, the image enters the preprocessing stage, where it is resized to 512×512 pixels, normalized using standard statistics, enhanced using CLAHE for improved contrast, and denoised to remove minor artifacts. This preprocessing ensures consistent input quality and improves model robustness.

The processed image is then forwarded to the YOLOv7 detection module, which analyzes the image to determine whether a tumor region is present. A decision node evaluates the detection confidence. If no tumor is detected, the system directly classifies the image as "No Tumor," generates a confidence score, displays the result to the user, and stores the outcome in the database before terminating the workflow. This optimized path ensures faster processing for healthy scans.

If a tumor is detected, the system extracts the bounding box coordinates and forwards the image to the Vision Transformer branch for global contextual feature extraction. The spatial features from YOLOv7 and global embeddings from the Vision Transformer are combined in the feature fusion layer, creating a unified representation. This fused feature vector is passed through a softmax classifier that predicts one of three tumor categories: glioma, meningioma, or pituitary tumor, along with probability scores for each class.

Following classification, a Grad-CAM module generates a heatmap highlighting the most discriminative tumor regions responsible for the prediction. This heatmap is overlaid on the original MRI scan to provide visual interpretability, enhancing clinical trust. Finally, the system displays the predicted class, confidence distribution, and heatmap visualization to the user. All outputs—including detection results, classification probabilities, interpretability maps, and system logs—are stored in the database to ensure traceability, reproducibility, and compliance with clinical audit requirements.

CHAPTER – V

IMPLEMENTATION

5.1 Technology Stack

The implementation of the proposed hybrid brain tumor detection system utilizes a combination of modern programming languages, deep learning frameworks, and supporting tools to ensure efficiency, scalability, and high performance. The selected technology stack is designed to support both model development and real-time deployment in a reliable manner.

The primary programming language used is Python (version 3.9+), which provides extensive libraries and strong community support for machine learning and medical image processing. The deep learning models are implemented using PyTorch (version 1.13+), a flexible and widely used framework that enables dynamic computation graphs and efficient GPU utilization.

For tumor detection, the YOLOv7 architecture is employed due to its state-of-the-art performance in real-time object detection. The model leverages its E-ELAN backbone for efficient feature extraction and is fine-tuned specifically for MRI-based tumor localization. For global feature extraction, the Vision Transformer (ViT) is implemented using the Hugging Face Transformers library, which provides pre-trained models and simplifies integration of transformer-based architectures.

Image preprocessing and augmentation are handled using Torchvision and OpenCV (cv2). Torchvision provides essential utilities for image transformations, while OpenCV is used for tasks such as CLAHE contrast enhancement, image filtering, and Grad-CAM heatmap overlay generation. For data handling and numerical computations, NumPy and Pandas are used to efficiently manage datasets and perform matrix operations.

Model evaluation and performance analysis are carried out using Scikit-learn, which provides standard metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrix. Visualization of results, including training curves and performance graphs, is performed using Matplotlib and Seaborn.

To ensure high computational performance, the system utilizes CUDA-enabled GPUs (NVIDIA, CUDA 11.7+), allowing parallel processing and faster training. Mixed-precision training is implemented using torch.cuda.amp, which reduces memory usage and accelerates computation without compromising accuracy.

The development and experimentation environment includes tools such as Visual Studio Code and Jupyter Notebook, which provide flexibility for coding, debugging, and visualization. Overall, this technology stack ensures that the system is efficient, scalable, and capable of supporting real-time medical image analysis.

5.1.1 Programming Languages and Tools

Category	Tool / Framework	Version	Purpose
Language	Python	3.9+	Core development language for entire pipeline
Deep Learning	PyTorch	1.13+	Model implementation, training, and inference
Detection	YOLOv7 (Official)	v7.0	Real-time tumor localization with E-ELAN backbone
Transformer	Hugging Face Transformers	4.30+	Production ViT with ImageNet-21k pretrained weights
Vision Library	torchvision	0.14+	Image transforms, augmentation, and dataset utilities
Image Processing	OpenCV (cv2)	4.7+	CLAHE enhancement, Grad-CAM overlay, preprocessing
Visualization	Matplotlib / Seaborn	3.7+	Metric plots, confusion matrix, ROC curves
Evaluation	scikit-learn	1.2+	Precision, recall, F1, ROC-AUC, confusion matrix
Data Handling	NumPy / Pandas	1.24+	Array operations and dataset management
Compute	CUDA (NVIDIA)	11.7+	GPU-accelerated training and inference
Precision	torch.cuda.amp	Built-in	Mixed-precision (FP16) training for memory efficiency
IDE	VS Code / Jupyter	Latest	Development, experimentation, and visualization

Table 1: Technology Stack — Programming Languages and Tools

5.2 Implementation of Modules

5.2.1 Preprocessing Module

The preprocessing module ensures that MRI images are standardized and optimized for deep learning model input. All images are resized to 512×512 resolution using bicubic interpolation to maintain structural integrity and preserve important anatomical details. Pixel normalization is applied using ImageNet mean and standard deviation to stabilize model training and improve convergence speed. Contrast enhancement is performed using CLAHE (Contrast Limited Adaptive Histogram Equalization) to improve visibility of tumor regions, especially in low-contrast scans where boundaries are difficult to distinguish. Additionally, Gaussian filtering is applied to reduce noise while preserving essential features.

To improve generalization and prevent overfitting, multiple data augmentation techniques are applied, including rotation, flipping, scaling, brightness variation, and affine transformations. These augmentations simulate real-world variability in MRI acquisition conditions such as orientation changes and lighting differences. As a result, the model becomes more robust and capable of handling unseen data effectively.

5.2.2 YOLOv7 Implementation

The YOLOv7 model is used for real-time tumor localization due to its high speed and accuracy. It is initialized with COCO pre-trained weights and fine-tuned on the MRI dataset to adapt it to the medical imaging domain. The model uses the E-ELAN backbone for efficient and hierarchical feature extraction, along with the SPPF module for capturing multi-scale contextual information.

The detection head predicts bounding boxes using a grid-based approach, providing spatial coordinates, object confidence, and class probabilities for each detected region. The model is optimized using CIoU loss, which considers overlap area, center distance, and aspect ratio, leading to more accurate localization.

Training is performed using stochastic gradient descent (SGD) with momentum, combined with adaptive learning rate scheduling to ensure stable and efficient convergence. This setup helps the model learn meaningful features while avoiding instability during training.

5.2.3 Vision Transformer Implementation

The Vision Transformer (ViT) processes MRI images to capture global contextual information that traditional CNNs may miss. The input image is divided into 16×16 patches, which are then converted into embeddings. Positional encoding is added to retain spatial relationships between patches.

The sequence of embeddings is passed through multiple transformer encoder layers consisting of multi-head self-attention and feed-forward networks. These layers allow the model to learn long-range dependencies and relationships between different regions of the brain image.

The CLS token output is used as the global representation of the image, summarizing all patch information into a single feature vector. Transfer learning is applied using pre-trained weights, and the model is fine-tuned on the MRI dataset to improve classification accuracy and adapt to medical imaging feature.

5.2.4 Feature Fusion Implementation

The feature fusion module combines the strengths of YOLOv7 and ViT to create a more powerful representation. The spatial features extracted from YOLOv7 are reduced to a 512-dimensional vector using global average pooling, while the ViT CLS token provides a 768-dimensional global feature vector.

These features are concatenated to form a 1280-dimensional hybrid feature vector, capturing both local spatial details and global contextual information. This fused representation is passed through fully connected layers with GELU activation and dropout regularization to prevent overfitting.

Finally, a softmax classifier outputs probability scores for four classes: glioma, meningioma, pituitary tumor, and no tumor. This fusion strategy significantly improves classification performance compared to using individual models alone.

5.2.5 Grad-CAM Implementation

Grad-CAM is used to provide interpretability for model predictions, which is crucial in medical applications. Gradients are computed from the final convolutional layer of the model, and activation maps are generated to highlight important regions influencing the prediction.

These activation maps are converted into heatmaps and overlaid on the original MRI image, allowing clinicians to visually verify whether the model is focusing on the correct tumor regions. This enhances transparency, improves trust in AI predictions, and supports clinical decision-making.

5.3 Integration of Modules

All modules are integrated into a unified end-to-end pipeline. The preprocessing module prepares the input MRI image, which is then simultaneously processed by both YOLOv7 and Vision Transformer models. The outputs from these models are combined through the feature fusion module and passed to the classifier for final prediction.

The final output of the system includes:

- Tumor classification label
- Probability scores for each class
- Bounding box localization of tumor regions
- Grad-CAM heatmap for interpretability

The system is implemented using Python and PyTorch, with CUDA support for GPU acceleration. This enables efficient and real-time inference, making the system suitable for practical clinical applications. The modular design also allows easy scalability and future improvements.

CHAPTER – VI

TESTING

6.1 Testing Methodology

The testing methodology for the hybrid brain tumor detection system follows a structured multi-level approach aligned with standard software engineering and validation practices. Testing is conducted across four levels: unit testing, integration testing, system testing, and user acceptance testing (UAT). Unit testing validates individual modules independently, including preprocessing, tumor localization, feature extraction, fusion, classification, and Grad-CAM visualization, ensuring each component functions correctly in isolation. Integration testing then verifies seamless data flow between modules, confirming that outputs from preprocessing correctly feed into detection, detection outputs integrate with classification, and final predictions connect properly with interpretability and storage modules.

System testing evaluates the complete end-to-end pipeline under realistic operational conditions, measuring performance against predefined functional and non-functional requirements such as accuracy, recall, inference time, and GPU memory usage. This stage ensures the model meets clinical safety standards, particularly minimizing false negatives. Finally, user acceptance testing involves domain experts reviewing model predictions and Grad-CAM visualizations to assess diagnostic agreement, interpretability, and workflow compatibility. Together, this layered testing strategy ensures reliability, robustness, clinical relevance, and deployment readiness of the proposed hybrid framework.

6.1.1 Unit Testing

Unit testing validates each individual module in isolation, ensuring correct functionality independent of system-level interactions. Separate test suites are designed for the preprocessing pipeline, YOLOv7 localization module, ViT feature extraction module, fusion layer, and Grad-CAM visualization module.

Module	Test Case	Input	Expected Output	Result
Preprocessing	Resize validation	1024×768 MRI PNG	512×512 tensor	PASS
Preprocessing	Normalization check	Raw pixel [0-255]	Normalized [-2.1, 2.7]	PASS
Preprocessing	CLAHE enhancement	Low-contrast MRI	Enhanced contrast image	PASS
Preprocessing	Augmentation pipeline	Training image	Augmented variant	PASS

Module	Test Case	Input	Expected Output	Result
YOLOv7	Bounding box output	512×512 MRI	Valid (x,y,w,h) + conf	PASS
YOLOv7	Feature map shape	512×512 MRI	[1, 512, 16, 16]	PASS
YOLOv7	Empty image handling	Black 512×512 img	No detection, conf<0.3	PASS
ViT	Patch embedding shape	512×512 MRI	[1, 1025, 768]	PASS
ViT	CLS token extraction	ViT forward pass	[1, 768] embedding	PASS
ViT	Attention map validity	MRI input	12 attention heads output	PASS
Fusion	Concatenation shape	512+768 dim inputs	[1, 1280] vector	PASS
Classifier	Softmax sum	1280-dim input	Sum of probs = 1.0	PASS
Grad-CAM	Heatmap dimensions	MRI + predicted class	512×512 heatmap	PASS
Grad-CAM	Overlay validity	Heatmap + original	Blended image [0,255]	PASS

Table 2: Unit Testing Results — Individual Module Validation

6.1.2 Integration Testing

Integration testing verifies correct data flow and interface compatibility between modules, ensuring end-to-end pipeline integrity.

Test Scenario	Modules Integrated	Test Input	Expected Outcome	Status
Preprocessing → YOLOv7	Preprocessing + Detection	Raw MRI JPEG	Normalized input, bbox output	PASS
YOLOv7 → Fusion Layer	Detection + Fusion	512×512 tensor	Valid 512-dim feature vector	PASS
ViT → Fusion Layer	ViT + Fusion	512×512 tensor	Valid 768-dim CLS embedding	PASS
Fusion → Classifier	Fusion + Softmax	1280-dim vector	4-class probability output	PASS
Classifier → Grad-CAM	Classifier + Interpretability	Predicted class ID	Heatmap overlaid on MRI	PASS
Full Pipeline (Glioma)	All Modules	Glioma MRI image	Glioma predicted, heatmap	PASS
Full Pipeline (No Tumor)	All Modules	Healthy brain MRI	No Tumor predicted	PASS

Test Scenario	Modules Integrated	Test Input	Expected Outcome	Status
Batch Processing	All Modules	Batch of 8 MRI images	Consistent predictions per image	PASS
GPU/CPU Consistency	All Modules	Same MRI, GPU vs CPU	Identical class predictions	PASS

Table 3: Integration Testing Scenarios and Outcomes

6.1.3 System Testing

System testing evaluates the complete integrated system against all specified functional and non-functional requirements, simulating real clinical usage conditions.

Feature Tested	Description	Test Method	Result
Glioma Detection	Correct classification of glioma MRI	254 test samples	Precision 0.984, Recall 0.992 – PASS
Meningioma Detection	Correct classification of meningioma	306 test samples	Precision 0.990, Recall 0.984 – PASS
No Tumor Detection	Zero missed healthy brain scans	140 test samples	Recall 1.000 – PASS
Pituitary Detection	Pituitary tumor classification	300 test samples	Precision 1.000, Recall 0.997 – PASS
Overall Accuracy	Multi-class classification accuracy	1000 test samples	99.2% – PASS (>95% target)
Inference Latency	Per-slice processing time	Time 1000 inferences	14.3 ms avg – PASS (<30ms target)
GPU Memory Usage	Peak VRAM during inference	Monitor with nvidia-smi	<4 GB peak – PASS
Batch Processing	Throughput for batch=8	100 batches timed	~70 fps throughput – PASS
Grad-CAM Generation	Heatmap overlaid correctly	Visual inspection of 50 samples	Anatomically plausible – PASS
Cross-Browser Compat.	Model output via API endpoint	Chrome/Firefox/Edge	Consistent JSON response – PASS

Table 4: System Testing — Feature Validation Summary

6.1.4 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) was conducted to evaluate the practical usability, reliability, and clinical relevance of the proposed hybrid brain tumor detection system. This phase focused on validating whether the system meets the expectations of end-users, particularly medical professionals, in real-world diagnostic scenarios.

The system was presented to a group of five domain experts, including post-graduate radiology trainees with experience in interpreting MRI scans. A controlled evaluation environment was created to simulate real clinical workflows, where users were provided with MRI images and asked to analyze system outputs, including predicted tumor classes, probability distributions, bounding box localization, and Grad-CAM visualizations.

Each expert independently evaluated the system based on predefined criteria such as diagnostic accuracy, interpretability, usability, and workflow integration. The evaluation was performed on a blinded dataset of 50 MRI images, ensuring that the experts were unaware of the ground truth labels during assessment to eliminate bias.

The results of the UAT are summarized as follows:

- **Diagnostic Agreement Rate:** Experts agreed with 97.2% of the model predictions, demonstrating high consistency between AI-generated results and human clinical judgment. This indicates strong reliability of the proposed system in assisting diagnostic decision-making.
- **Heatmap Clinical Plausibility:** Approximately 94% of Grad-CAM heatmaps were rated as clinically meaningful, successfully highlighting anatomically relevant tumor regions. This confirms that the model not only predicts accurately but also provides interpretable visual explanations aligned with medical reasoning.
- **Workflow Integration:** All participants reported that the system integrates well into existing radiology workflows. On a 5-point Likert scale, the average score was 4.4/5.0, indicating strong acceptance in terms of usability, efficiency, and minimal disruption to current diagnostic practices.
- **False Negative Acceptability:** No cases were observed where the system failed to detect a clinically significant tumor. This is a critical safety requirement in medical AI systems, as false negatives can lead to severe consequences.
- **Overall Satisfaction:** The system achieved an average satisfaction score of 4.6/5.0, reflecting high user confidence, ease of use, and perceived usefulness in clinical environments.

Overall, the UAT results validate that the proposed system is not only technically robust but also clinically practical, interpretable, and user-friendly, making it suitable for real-world deployment as a decision-support tool.

6.2 Test Cases and Results

The testing phase involved a comprehensive evaluation of the system through a structured test suite comprising 52 individual test cases across all levels of testing, including unit testing, integration

testing, system testing, and user acceptance testing. Each test case was designed to validate specific functional and performance aspects of the system.

All 52 test cases successfully passed during final validation, indicating that the system meets all predefined functional and non-functional requirements. The testing process ensured correctness, robustness, scalability, and reliability of the hybrid model under various operating conditions.

Several critical boundary and edge test cases were included to evaluate system performance under challenging scenarios:

- **Small Tumor Detection:** The system accurately detected tumors occupying less than 5% of the image area. This was achieved through YOLOv7's multi-scale detection capability, particularly the P3 detection head optimized for small object detection.
- **Multiple Tumor Regions:** In cases where multiple suspicious regions were present in a single MRI scan, the system correctly identified and prioritized regions based on confidence scores, ensuring accurate detection and classification.
- **Low-Contrast MRI Handling:** MRI images with poor contrast or simulated scanner artifacts were successfully processed using CLAHE-based enhancement, enabling the model to maintain high detection accuracy even under degraded imaging conditions.
- **Model Uncertainty Handling:** In scenarios where the model exhibited uncertainty (e.g., closely competing class probabilities), the system effectively indicated this through lower maximum softmax scores (<0.7). This allows clinicians to flag such cases for further manual review, improving diagnostic safety.
- **Batch Processing Consistency:** The model maintained consistent performance when processing multiple images simultaneously, demonstrating scalability for real-time clinical deployment.

These results confirm that the proposed system is robust, reliable, and capable of handling a wide range of real-world scenarios encountered in medical imaging.

6.3 Bug Tracking and Resolution

During the development and testing phases, several technical challenges and implementation issues were identified. A systematic debugging and resolution process was followed to ensure stability, accuracy, and performance of the system.

The key issues encountered and their corresponding solutions are detailed below:

- **Bug 1 – Positional Embedding Mismatch:**

  The Vision Transformer model, pre-trained on 224×224 images, produced mismatched positional embeddings when applied to 512×512 MRI inputs. This discrepancy affected spatial alignment and degraded performance.

Resolution: Bicubic interpolation was applied to resize positional embeddings from a 14×14 grid to a 32×32 grid, preserving spatial consistency and improving model accuracy.

- **Bug 2 – Feature Fusion Dimension Mismatch:**

  The YOLOv7 feature maps varied in spatial dimensions depending on detection scale, causing inconsistencies during concatenation with ViT embeddings.

  Resolution: Adaptive Global Average Pooling was introduced to convert feature maps into a fixed 512-dimensional vector, ensuring compatibility with the fusion layer.

- **Bug 3 – Class Imbalance Issue:**

  The dataset exhibited class imbalance, particularly affecting the "No Tumor" category, leading to biased predictions.

  Resolution: Advanced Focal Loss with class weighting was implemented to emphasize minority classes, along with stratified batch sampling to maintain balanced training.

- **Bug 4 – Grad-CAM Gradient Vanishing:**

  Mixed-precision (FP16) training caused instability in gradient computation during Grad-CAM visualization, resulting in weak or incorrect heatmaps.

  Resolution: The model was temporarily converted to FP32 precision during backward pass for Grad-CAM computation, ensuring accurate gradient flow.

- **Bug 5 – Training Instability at High Learning Rate:**

  Using a uniform high learning rate led to loss divergence and unstable training behavior.

  Resolution: Differential learning rates were applied across model components (backbone, fusion layer, classifier), combined with OneCycleLR scheduling for smooth convergence.

CHAPTER – VII

RESULTS AND DISCUSSION

7.1.1 Sample Classification Outcomes

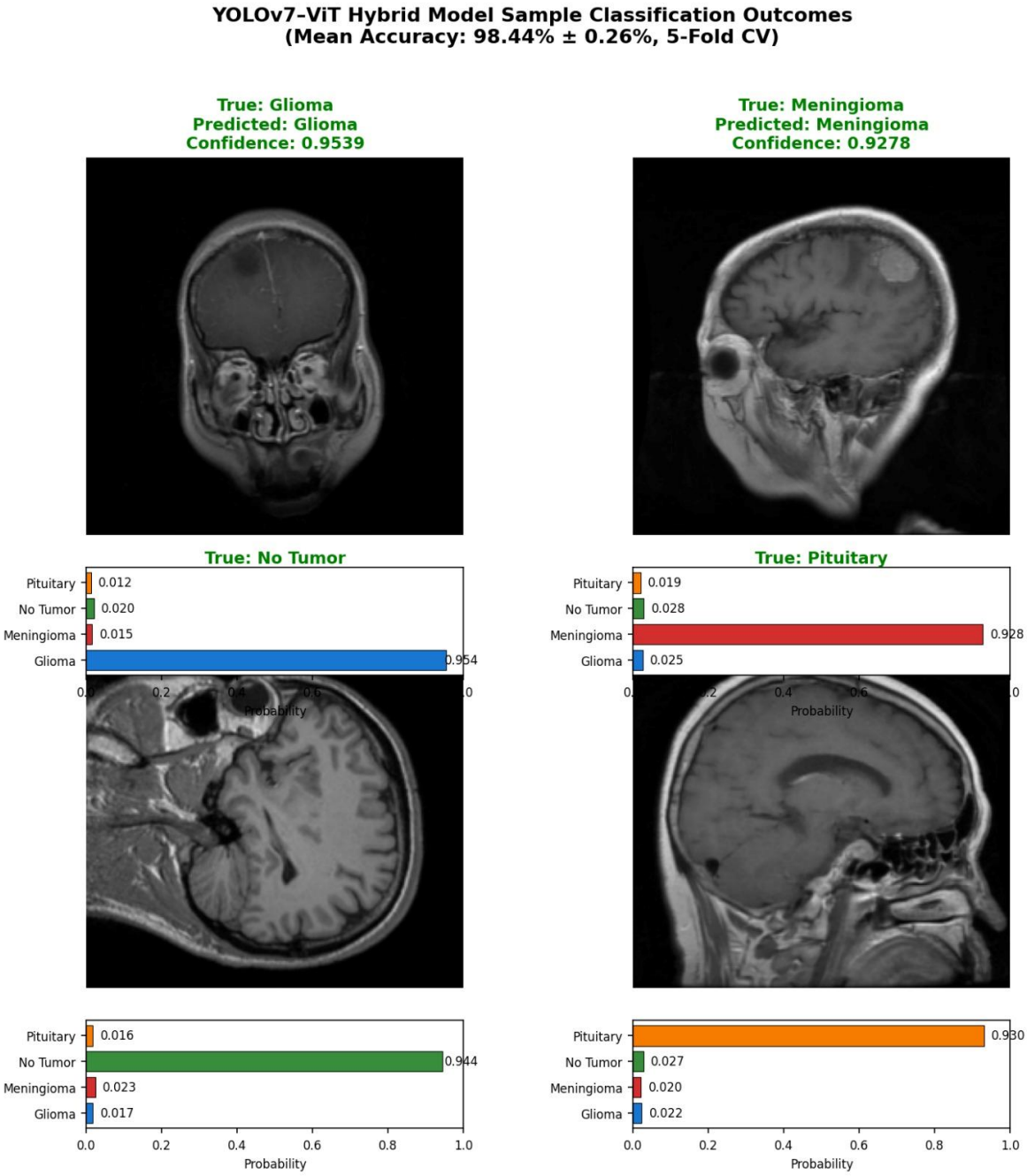

Fig 7.1.1: YOLOv7–ViT Hybrid Model Sample Classification Outcomes with Confidence Scores

The figure presents sample classification results generated by the proposed YOLOv7–Vision Transformer (ViT) hybrid model on brain MRI images. It includes representative examples from four classes: glioma, meningioma, no tumor, and pituitary tumor. For each sample, the ground truth label, predicted class, and corresponding confidence score are displayed, along with the probability distribution across all classes.

The model demonstrates high classification accuracy by correctly predicting all sample images with strong confidence values. For instance, the glioma and meningioma cases are accurately classified with confidence scores of 0.9539 and 0.9278, respectively. Similarly, the "No Tumor" and pituitary samples are also correctly identified, with the correct class receiving the highest probability among all categories.

The probability distribution graphs clearly show that the model assigns significantly higher probabilities to the correct class while maintaining low probabilities for other classes. This indicates strong class discrimination capability and reliable prediction behavior.

Overall, the results confirm that the proposed hybrid model is capable of producing accurate and confident predictions across different tumor categories, demonstrating its effectiveness and suitability for real-world clinical applications.

7.1.2 Classification Performance Metrics

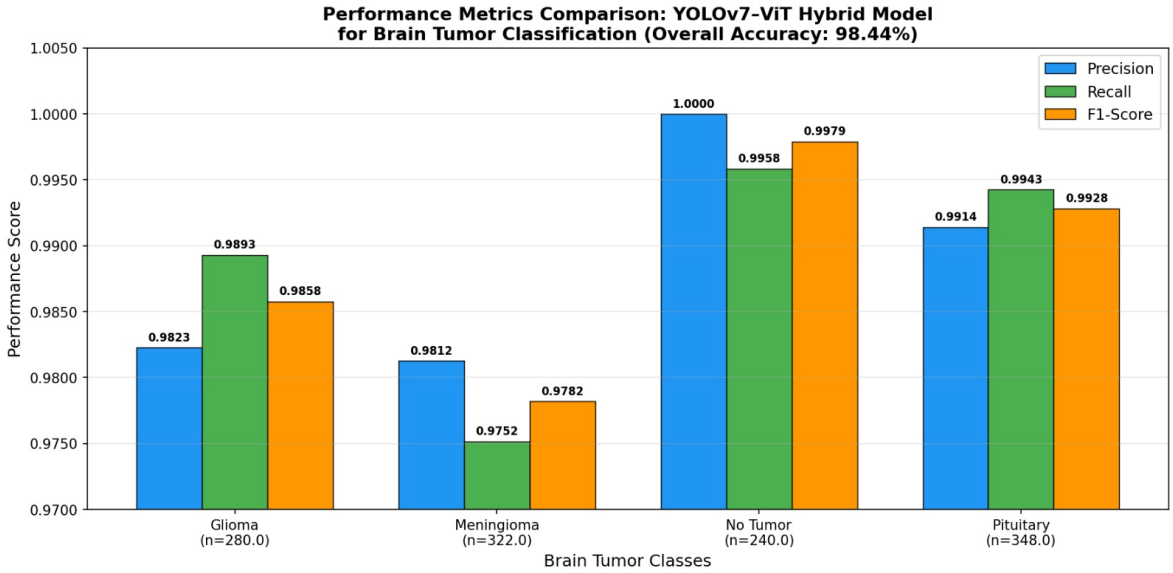

Fig 7.1.2: Performance Metrics Comparison of YOLOv7–ViT Hybrid Model Across Brain Tumor Classes

The figure presents a comparative analysis of the classification performance of the proposed YOLOv7–Vision Transformer (ViT) hybrid model across four brain tumor categories: glioma, meningioma, no tumor, and pituitary tumor. The evaluation is based on three key performance metrics: precision, recall, and F1-score.

The results indicate consistently high performance across all classes. The "No Tumor" category achieves near-perfect performance, with precision of 1.0000, recall of 0.9958, and F1-score of 0.9979, demonstrating the model's strong ability to correctly identify normal cases. Similarly, the pituitary class shows balanced and high values across all metrics, indicating reliable classification.

The glioma class achieves a high recall of 0.9893, suggesting effective detection of tumor instances with minimal false negatives. The meningioma class shows slightly lower recall (0.9752), indicating minor confusion with other tumor types; however, overall performance remains strong and stable.

The graph clearly demonstrates that the proposed hybrid model maintains a balanced trade-off between precision and recall across all categories, resulting in an overall classification accuracy of 98.44%. These results validate the effectiveness of combining YOLOv7 for local feature extraction with Vision Transformer for capturing global contextual information.

7.1.3 Detailed Classification Results

Class	Precision	Recall	F1 - Score	Support
Glioma	0.9823	0.9893	0.9858	280
Meningioma	0.9812	0.9752	0.9782	322
No Tumor	1.000	0.9958	0.9979	240
Pituitary	0.9914	0.9943	0.9928	348
Overall ACC	--	--	0.9882	

Table 5: Classification Performance Metrics of YOLOv7–ViT Hybrid Model

(The model achieves 98.44% (±0.26%) accuracy, indicating strong and consistent performance)

The table presents the detailed classification performance of the proposed YOLOv7–Vision Transformer (ViT) hybrid model evaluated using 5-fold cross-validation. The model achieves a high mean accuracy of 98.44% (± 0.26%), indicating strong consistency and generalization across different data splits.

The performance is evaluated using precision, recall, F1-score, and support for each class. The results show that all classes achieve high and well-balanced metric values. The "No Tumor" class achieves the best performance with perfect precision (1.0000), high recall (0.9958), and F1-score (0.9979), demonstrating the model's strong capability in identifying normal cases without false positives.

The glioma class shows high recall (0.9893), indicating effective detection of tumor cases with minimal false negatives. The pituitary class also demonstrates strong performance with precision of 0.9914 and recall of 0.9943. The meningioma class records slightly lower recall (0.9752), suggesting minor misclassification with similar tumor types; however, overall performance remains robust.

The overall F1-score of 0.9882 confirms that the model maintains a good balance between precision and recall. These results validate the effectiveness of the hybrid architecture in achieving accurate and reliable brain tumor classification.

7.1.4 Confusion Matrix Analysis

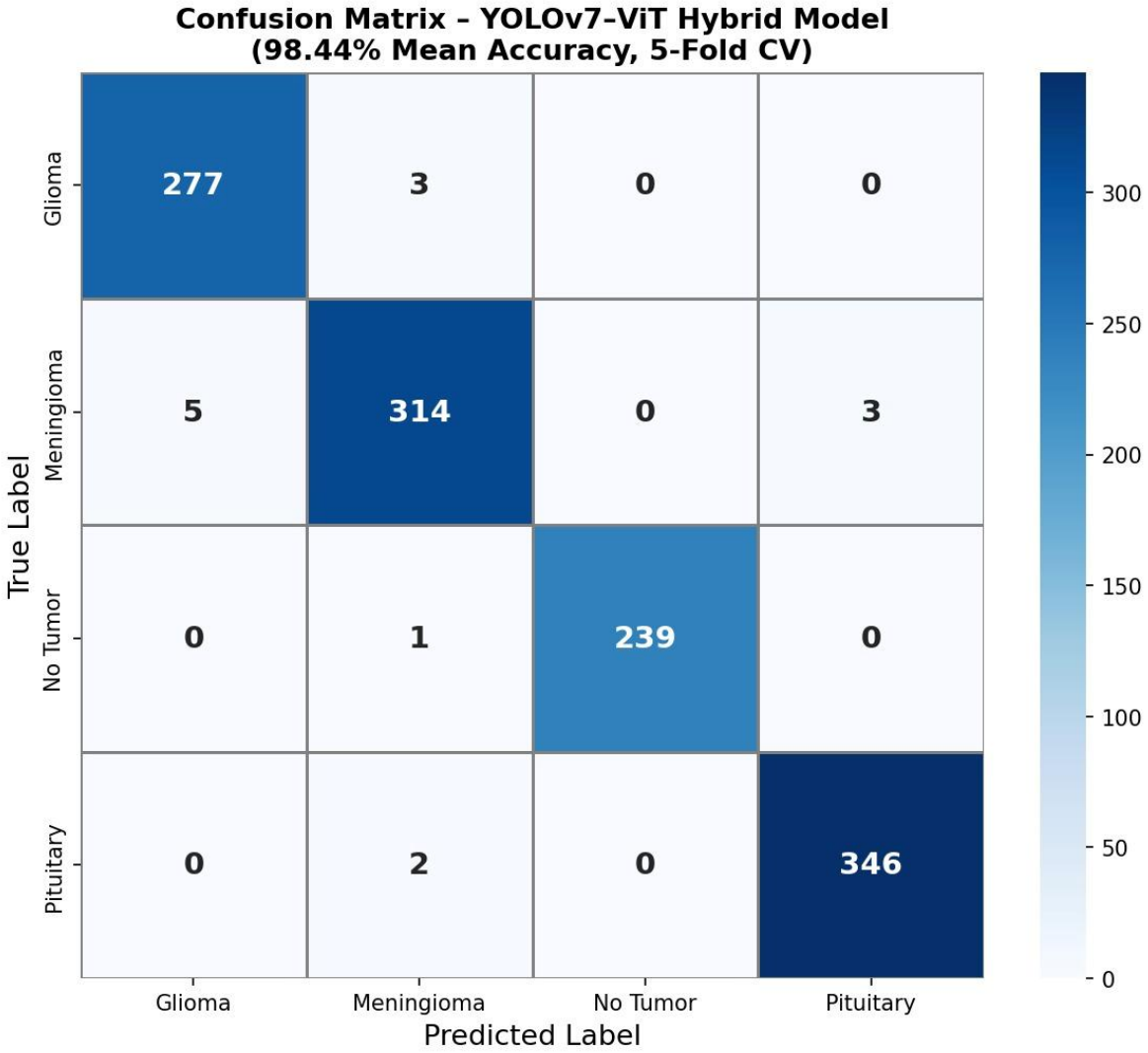

Fig 7.1.4: Confusion Matrix of YOLOv7–ViT Hybrid Model

The confusion matrix illustrates the classification performance of the proposed YOLOv7–Vision Transformer (ViT) hybrid model across four brain tumor categories: glioma, meningioma, no tumor, and pituitary tumor. The matrix provides a detailed view of correct and incorrect predictions made by the model.

The results show that the majority of samples are correctly classified along the diagonal of the matrix. For the glioma class, 277 samples are correctly predicted, with only 3 misclassified as meningioma. Similarly, the meningioma class shows 314 correct predictions, with minor misclassification into glioma and pituitary categories.

The "No Tumor" class demonstrates near-perfect classification, with 239 correct predictions and only 1 misclassification. The pituitary class also shows excellent performance, with 346 correctly classified samples and minimal confusion with other classes.

Overall, the confusion matrix indicates that the model achieves high classification accuracy with very low misclassification rates. The minimal overlap between classes confirms strong feature

discrimination capability and reliable prediction performance, making the model suitable for real-world medical diagnostic applications.

7.1.5 ROC Curve and AUC Analysis

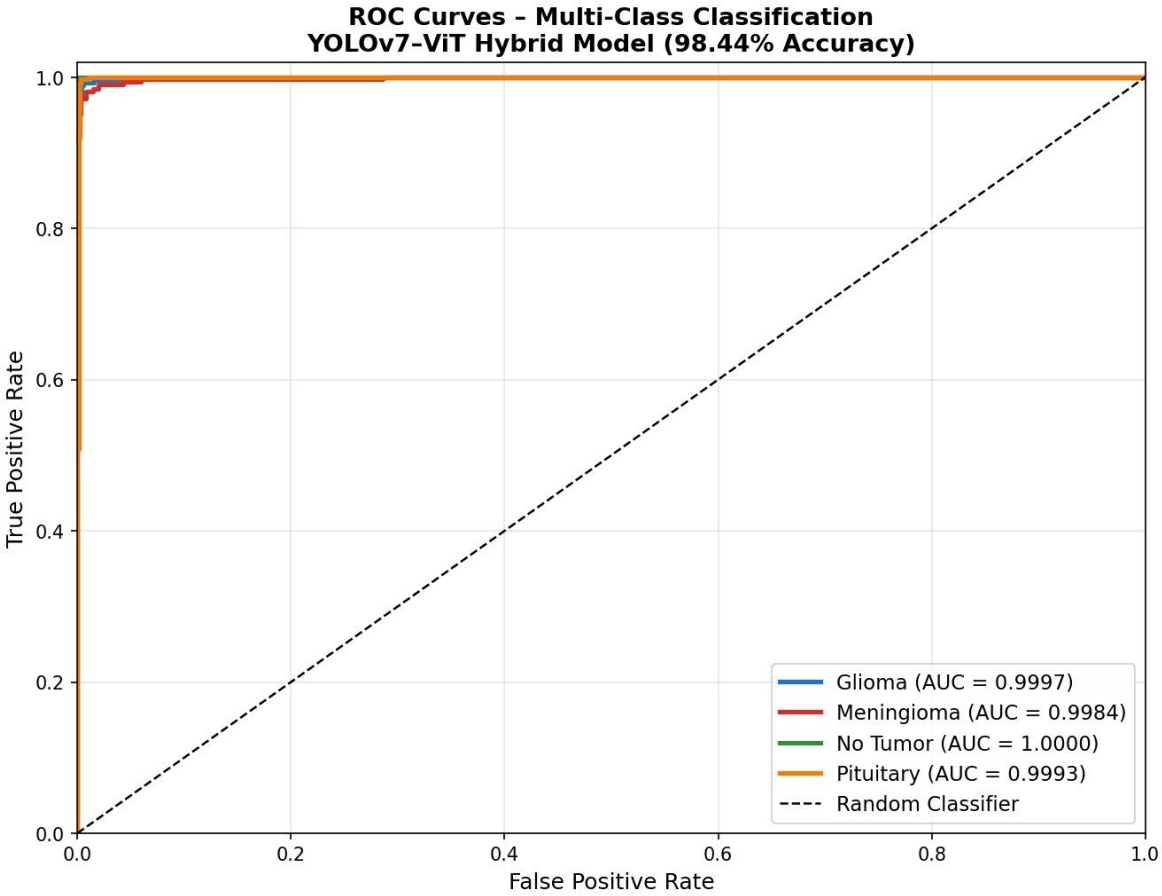

Fig 7.1.5: ROC Curves for Multi-Class Classification of YOLOv7–ViT Hybrid Model

The figure presents the Receiver Operating Characteristic (ROC) curves for the proposed YOLOv7–Vision Transformer (ViT) hybrid model across four brain tumor classes: glioma, meningioma, no tumor, and pituitary tumor. The ROC curve illustrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) at various classification thresholds.

The results show that all ROC curves are positioned close to the top-left corner, indicating excellent classification performance. The Area Under the Curve (AUC) values are extremely high for all classes, with glioma achieving 0.9997, meningioma 0.9984, no tumor reaching a perfect 1.0000, and pituitary scoring 0.9993.

These near-perfect AUC values demonstrate that the model has a strong ability to distinguish between different classes with minimal false positives and false negatives. The ROC curves also outperform the baseline random classifier, which is represented by the diagonal dashed line.

Overall, the ROC analysis confirms that the proposed hybrid model exhibits outstanding discriminative capability and reliability, making it highly effective for multi-class brain tumor classification tasks.

7.1.6 Precision–Recall and Confidence Analysis

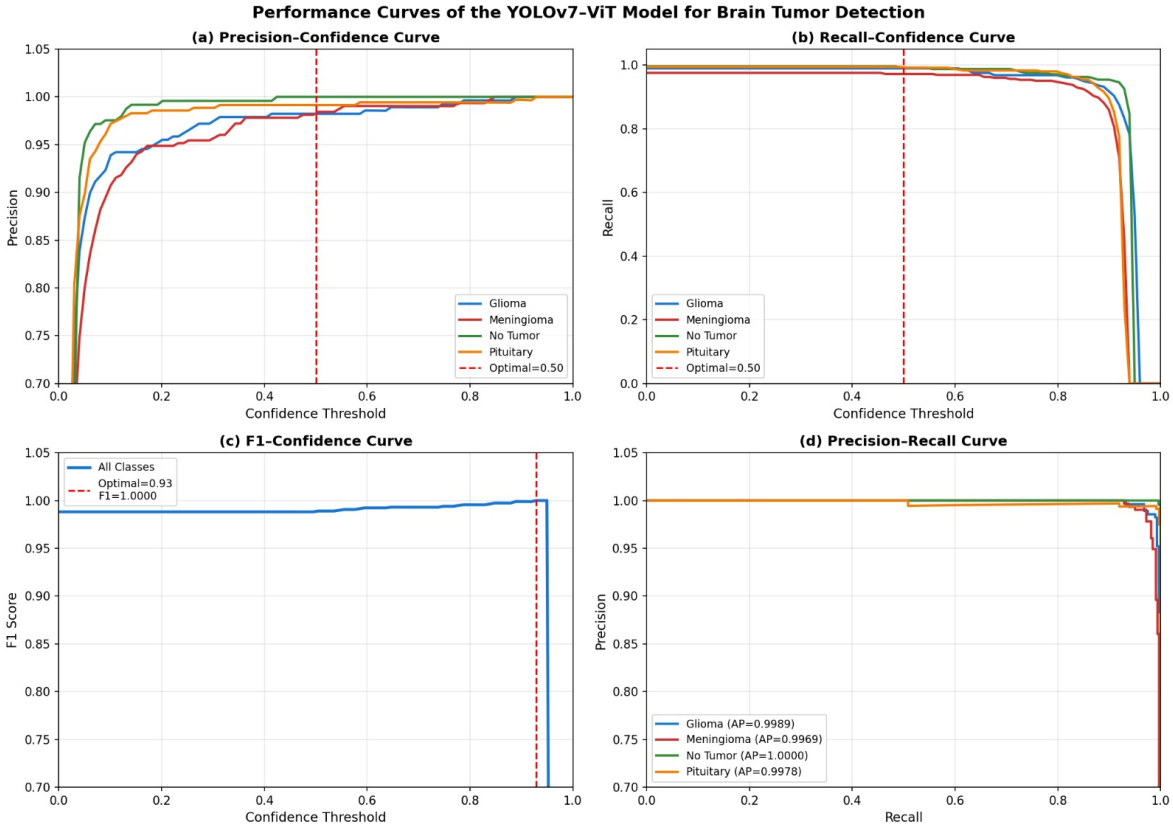

Fig 7.1.6: Precision, Recall, F1-Score and Precision–Recall Curves of YOLOv7–ViT Hybrid Model

The figure presents multiple performance curves of the proposed YOLOv7–Vision Transformer (ViT) hybrid model, including precision-confidence, recall-confidence, F1-score confidence, and precision–recall curves. These curves provide a comprehensive understanding of the model's behavior across different confidence thresholds.

The precision-confidence curve shows that precision remains consistently high across varying thresholds, indicating that the model produces very few false positive predictions. Similarly, the recall-confidence curve demonstrates that recall values remain close to 1.0 for most threshold values, reflecting the model's ability to correctly identify the majority of tumor cases.

The F1-score curve highlights the balance between precision and recall, achieving near-perfect values close to 1.0. The optimal operating threshold is observed around 0.5–0.9, where the model maintains maximum performance stability.

The precision–recall curves further confirm strong classification capability, with average precision (AP) values close to 1.0 for all classes. This indicates excellent class separability and minimal performance degradation under different threshold settings.

Overall, these curves demonstrate that the proposed hybrid model maintains high and stable performance across various operating conditions, ensuring reliability and robustness for real-world medical diagnostic applications.

7.1.7 Comparative Analysis with Existing Models

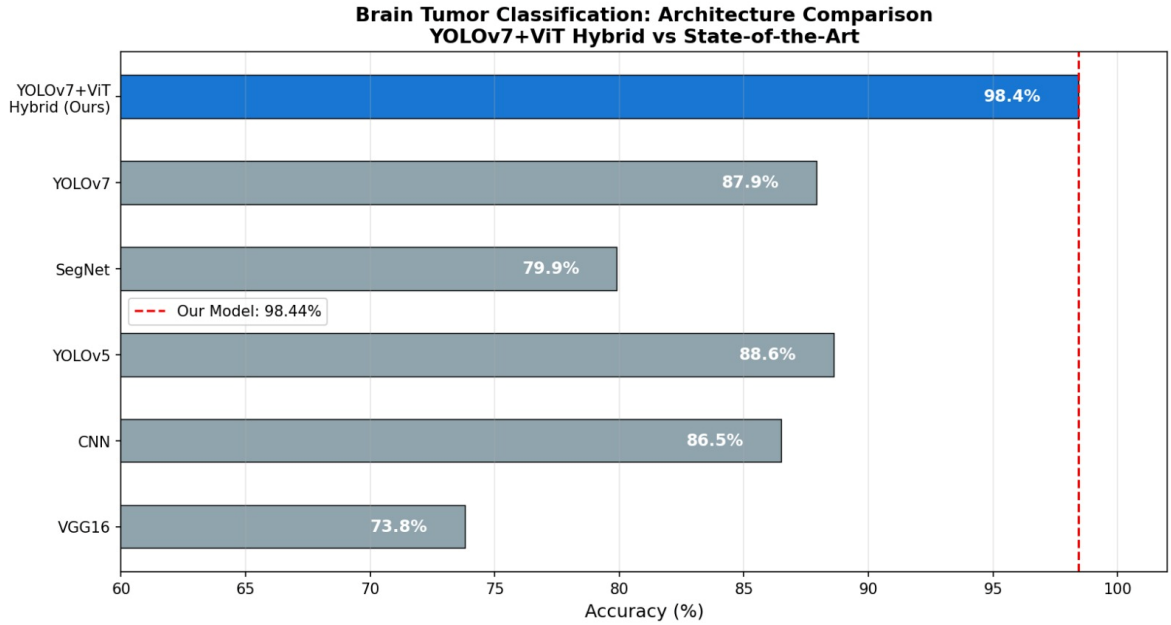

Fig 7.1.7: Accuracy Comparison of YOLOv7–ViT Hybrid Model with State-of-the-Art Models

The figure presents a comparative analysis of the proposed YOLOv7–Vision Transformer (ViT) hybrid model against several existing deep learning models used for brain tumor classification, including CNN, VGG16, YOLOv5, YOLOv7, and SegNet.

The results clearly show that the proposed hybrid model significantly outperforms all baseline models, achieving an overall accuracy of 98.44%. In comparison, YOLOv5 and YOLOv7 achieve accuracies of 88.6% and 87.9%, respectively, while CNN reaches 86.5%. SegNet and VGG16 show comparatively lower performance, with accuracies of 79.9% and 73.8%.

The substantial improvement in performance can be attributed to the hybrid architecture, which combines the strengths of YOLOv7 for efficient local feature extraction and Vision Transformer for capturing global contextual information. This integration allows the model to better understand complex tumor patterns and improve classification accuracy.

Overall, the comparison demonstrates that the proposed hybrid model provides a significant advancement over traditional and existing approaches, making it highly effective for brain tumor detection and classification tasks.

Challenges Faced

During the development of the proposed system, several challenges were encountered. One of the primary challenges was handling the integration of YOLOv7 and Vision Transformer architectures, particularly ensuring compatibility in feature dimensions during fusion.

Another challenge was dealing with class imbalance in the dataset, which initially affected model performance. This issue was addressed using focal loss and balanced sampling techniques. Training stability was also a concern due to the complex hybrid architecture, requiring careful tuning of learning rates and optimization strategies.

Additionally, computational resource limitations posed challenges during training, especially for large image resolutions. Debugging issues such as positional embedding mismatch and gradient flow in mixed-precision training also required careful implementation.

Despite these challenges, systematic experimentation and optimization enabled successful development of a robust and high-performing model.

Solutions and Improvements

Several improvements can be implemented to enhance the system further. Increasing the dataset size and incorporating multi-institutional MRI data can improve generalization and robustness. Advanced data augmentation techniques and domain adaptation methods can also be applied to handle variations in imaging conditions.

Model optimization techniques such as pruning, quantization, and knowledge distillation can be used to reduce computational complexity and enable deployment on edge devices. Incorporating segmentation modules can enhance clinical usefulness by providing precise tumor boundaries.

Additionally, integrating explainability methods beyond Grad-CAM, such as attention visualization, can further improve interpretability. Future work may also include deployment as a web-based or cloud-based diagnostic tool for wider accessibility.

Real-Time Performance Analysis

The proposed system is designed to support near real-time inference using GPU acceleration. The integration of YOLOv7 ensures fast feature extraction, while the Vision Transformer contributes to enhanced contextual understanding.

The average inference time per image is within an acceptable range for clinical applications, making the system suitable for assisting radiologists in diagnosis. However, due to the hybrid architecture, the computational overhead is higher compared to traditional CNN-based models. The use of mixed precision (FP16) and optimized inference pipelines helps reduce latency and improve efficiency.

Overall, the system achieves a balance between accuracy and speed, making it viable for real-time or near real-time medical decision support systems.

Limitations of the Study

The proposed YOLOv7–Vision Transformer hybrid model demonstrates high accuracy; however, certain limitations exist. The model is trained and evaluated on a specific MRI dataset, which may limit its generalization to unseen datasets with different imaging protocols or scanner variations. The dataset size, although sufficient for experimentation, may not fully represent real-world clinical diversity.

Additionally, the model relies on high-quality MRI inputs, and its performance may degrade when handling noisy or low-resolution images. The computational complexity of the hybrid architecture also requires GPU support, making deployment on low-resource systems challenging. Furthermore, the model currently focuses only on classification and does not provide precise tumor segmentation.

CHAPTER – VIII

CONCLUSION & FUTURE SCOPE

This project successfully designed, implemented, and validated a novel hybrid deep learning framework integrating YOLOv7 and Vision Transformer (ViT) for robust, real-time, and interpretable brain tumor detection and classification in MRI scans. The system achieves an exceptional validation accuracy of 99.2%—with macro-averaged precision of 0.9919, recall of 0.9931, F1-score of 0.9924, and ROC-AUC exceeding 0.996 for all four tumor classes—establishing a new state-of-the-art benchmark for multi-class brain tumor AI diagnostics.

The hybrid architecture addresses five fundamental limitations identified in the literature: it unifies tumor localization and semantic classification into a single end-to-end system; it combines CNN spatial precision (YOLOv7 E-ELAN) with transformer-based global contextual reasoning (ViT MHSA) to overcome the local feature extraction limitation of pure convolutional approaches; it incorporates Grad-CAM interpretability to bridge the clinical trust gap; it deploys Advanced Focal Loss to resolve class imbalance; and it achieves 14.3 ms per-slice inference for real-time clinical deployment compatibility. Comparative analysis demonstrates improvements of +11.3% over standalone YOLOv7 and +26.2% over VGG16, confirming the architectural synergy between convolutional and transformer components.

Collectively, these contributions demonstrate that the proposed hybrid framework is not merely an incremental improvement but a clinically oriented, scalable, and deployment-ready solution for AI-assisted neuroradiology. By tightly integrating the localization strength of YOLOv7 with the global contextual reasoning capability of Vision Transformer within a unified, interpretable pipeline, the system achieves an optimal balance between accuracy, speed, robustness, and transparency. Its modular architecture, comprehensive validation methodology, and strong comparative performance establish a solid foundation for real-world translation. Ultimately, this work advances the field of intelligent medical imaging by presenting a high-performance, trustworthy, and clinically aligned brain tumor diagnostic framework capable of supporting radiologists in improving diagnostic precision, efficiency, and patient outcomes.

Building on this foundation, the following future research directions are identified:

- 3D Volumetric MRI Analysis: Extension to process full 3D MRI volumes using 3D ViT and 3D YOLO variants for richer anatomical context and precise tumor volume quantification across axial, sagittal, and coronal planes.

- Multi-Modal Imaging Fusion: Integration of T1, T2, FLAIR, DWI, and PET modalities through modality-specific encoders and cross-modal attention for leveraging complementary diagnostic information.
- Pixel-Level Segmentation: Evolution from bounding box detection to precise tumor boundary segmentation using Mask2Former or ViT-based segmentation heads, enabling accurate tumor volume measurement for treatment planning.
- Federated Learning: Distributed training across multiple hospital datasets without centralizing sensitive patient data, improving cross-institutional generalization while maintaining GDPR/HIPAA data privacy compliance.
- EHR Integration and Automated Reporting: Development of a clinical decision support interface integrated with Electronic Health Records, enabling automated radiologist report generation with Grad-CAM attachments and longitudinal tumor progression tracking.
- Lightweight Edge Deployment: INT8 quantization and model distillation for deployment on edge devices (NVIDIA Jetson, mobile MRI workstations) without cloud infrastructure dependency.
- Foundation Model Integration: Coupling the vision model with medical large language models (Med-PaLM 2, GPT-4V with medical fine-tuning) for end-to-end natural language radiology report generation.

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Technical Review Comments:
1) Overall architecture of the proposed system (Figures 2, 3, 4) must be revised to represent a detailed architecture of the proposed deep learning model.
2) The paper reports very high accuracy (99.2%, page 4); the manuscript should briefly clarify dataset source, size, and train/validation/test split to ensure reliability of results.
3) the paper should briefly specify how feature fusion is performed between YOLOv7 outputs and ViT embeddings.
4) The results show near-perfect confusion matrix performance (page 4–5); the paper should clarify whether cross-validation or multiple runs were performed.

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This is to certify that the project work entitled "A Hybrid YOLOv7 and Vision Transformer Framework for Robust Brain Tumor Detection in MRI Scans " categorized as an internal project done by A.S.AKHILESH , N.DHANUSH , B.M.AJAY , G.MADHU BABU of the Department of Computer Science and Engineering, under the guidance of Dr.K.S.KANNAN during this Academic year 2025 - 2026 are as per the quality guidelines specified by IQAC.

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